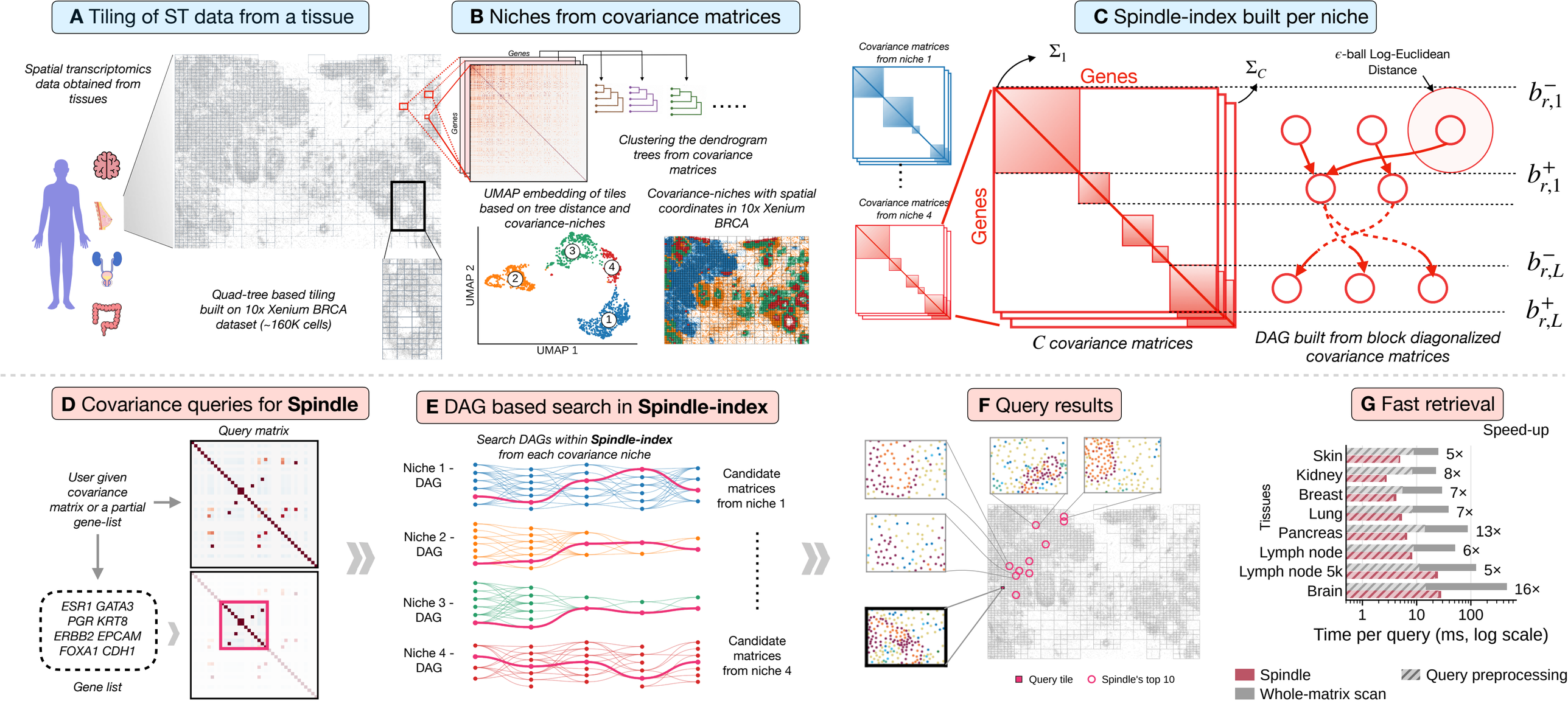

A Tiling of ST data from a tissue
Spatial transcriptomics data obtained from tissues
Quad-tree based tiling built on 10x Xenium BRCA dataset (~160K cells)
B Niches from covariance matrices
Genes
Genes
Clustering the dendrogram trees from covariance matrices
UMAP embedding of tiles based on tree distance and covariance-niches
UMAP 2
UMAP 1
1
2
3
4
Covariance-niches with spatial coordinates in 10x Xenium BRCA
C Spindle-index built per niche
Covariance matrices from niche 1
Covariance matrices from niche 4
Genes
Genes
Σ_1
Σ_C
ϵ-ball Log-Euclidean Distance
$b^-_{r,1}$
$b^+_{r,1}$
$b^-_{r,L}$
$b^+_{r,L}$
C covariance matrices
DAG built from block diagonalized covariance matrices
D Covariance queries for Spindle
Query matrix
User given covariance matrix or a partial gene-list
ESR1 GATA3
PGR KRT8
ERBB2 EPCAM
FOXA1 CDH1
Gene list
E DAG based search in Spindle-index
Search DAGs within Spindle-index from each covariance niche
Niche 1 - DAG
Niche 2 - DAG
Niche 3 - DAG
Niche 4 - DAG
Candidate matrices from niche 1
Candidate matrices from niche 4
F Query results
Query tile
Spindle's top 10
G Fast retrieval
Speed-up
Tissues
Skin 5×
Kidney 8×
Breast 7×
Lung 7×
Pancreas 13×
Lymph node 6×
Lymph node 5k 5×
Brain 16×
1
10
100
Time per query (ms, log scale)
Spindle
Whole-matrix scan
Query preprocessing